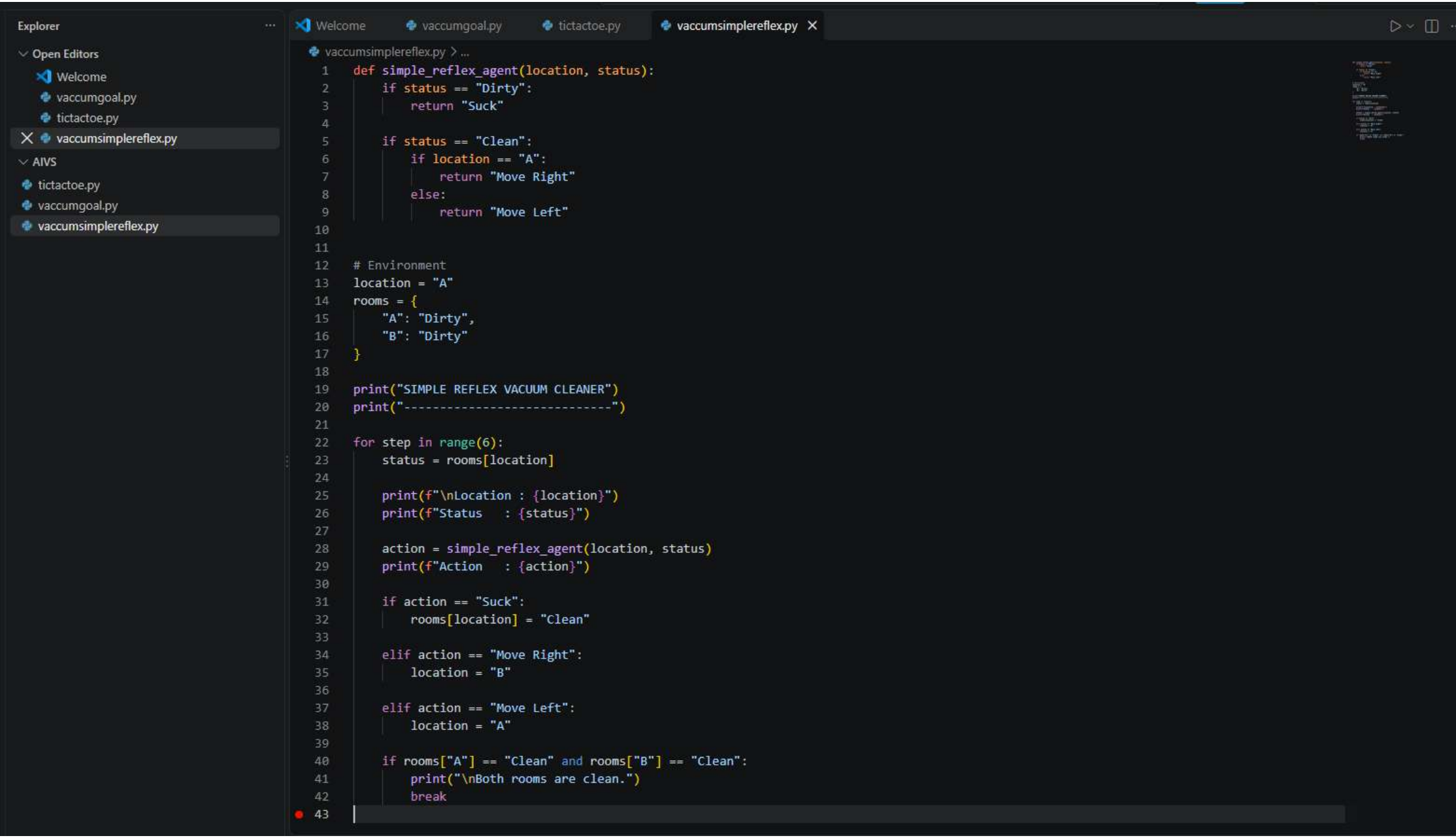

Explorer
Open Editors
Welcome
vaccumgoal.py
tictactoe.py
vaccumsimplereflex.py
AIVS
tictactoe.py
vaccumgoal.py
vaccumsimplereflex.py
Welcome
vaccumgoal.py
tictactoe.py
vaccumsimplereflex.py
vaccumsimplereflex.py > ...
1 def simple_reflex_agent(location, status):
2 if status == "Dirty":
3 return "Suck"
4
5 if status == "Clean":
6 if location == "A":
7 return "Move Right"
8 else:
9 return "Move Left"
10
11
12 # Environment
13 location = "A"
14 rooms = {
15 "A": "Dirty",
16 "B": "Dirty"
17 }
18
19 print("SIMPLE REFLEX VACUUM CLEANER")
20 print("----------")
21
22 for step in range(6):
23 status = rooms[location]
24
25 print(f"\nLocation : {location}")
26 print(f"Status : {status}")
27
28 action = simple_reflex_agent(location, status)
29 print(f"Action : {action}")
30
31 if action == "Suck":
32 rooms[location] = "Clean"
33
34 elif action == "Move Right":
35 location = "B"
36
37 elif action == "Move Left":
38 location = "A"
39
40 if rooms["A"] == "Clean" and rooms["B"] == "Clean":
41 print("\nBoth rooms are clean.")
42 break
43

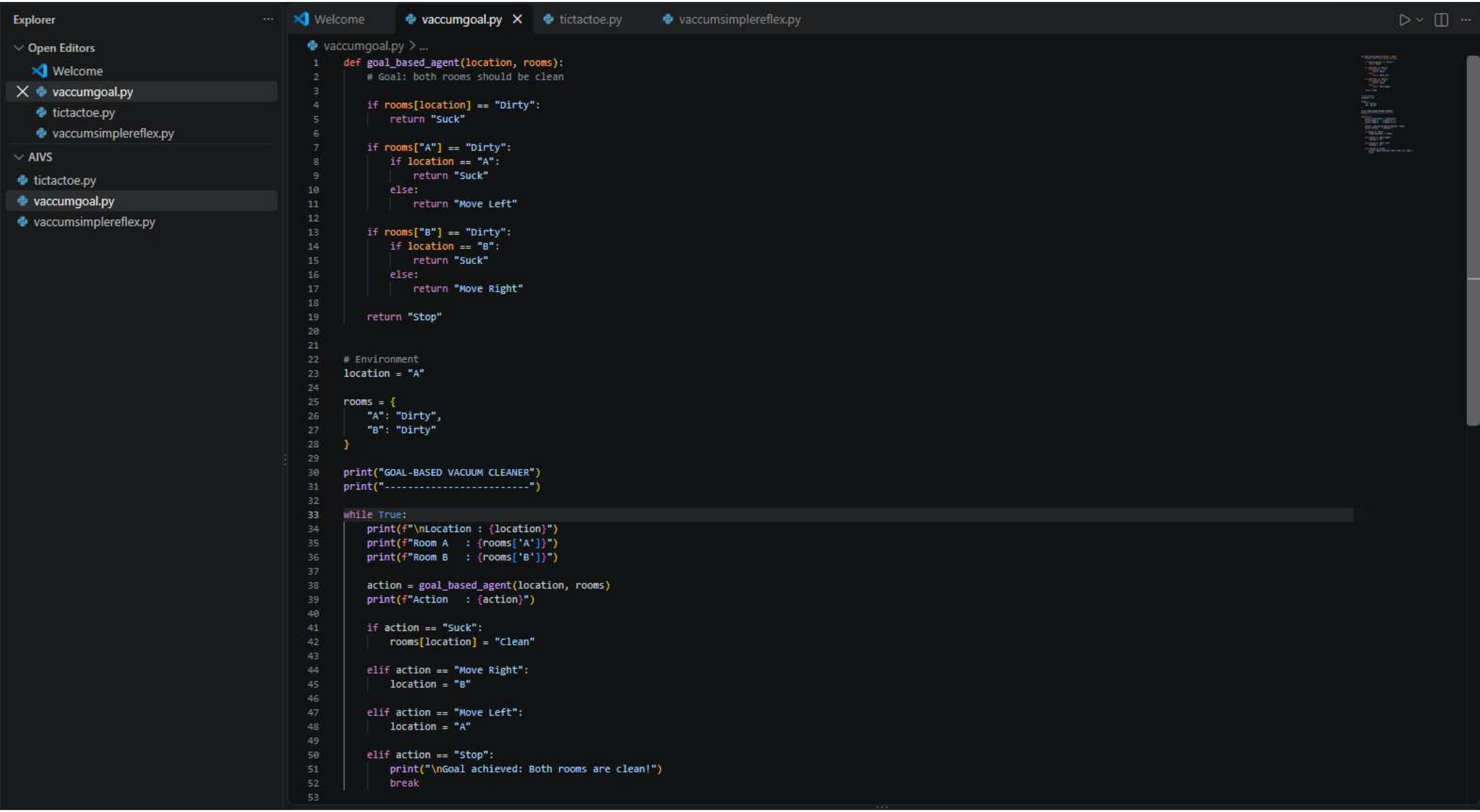
```
PS C:\Users\BMSCE\Desktop\AIVS>  & 'c:\python314\python.exe' 'c:\Users\BMSCE\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '49856' '--' 'c:\Users\BMSCE\Desktop\AIVS\vaccumsimplereflex.py'
SIMPLE REFLEX VACUUM CLEANER
----------------------------

Location : A
Status   : Dirty
Action   : Suck

Location : A
Status   : Clean
Action   : Move Right

Location : B
Status   : Dirty
Action   : Suck

Both rooms are clean.
```


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vaccumsimplereflex.py
vaccumgoal.py > ...
def goal_based_agent(location, rooms):
Goal: both rooms should be clean
if rooms[location] == "Dirty":
return "Suck"
if rooms["A"] == "Dirty":
if location == "A":
return "Suck"
else:
return "Move Left"
if rooms["B"] == "Dirty":
if location == "B":
return "Suck"
else:
return "Move Right"
return "Stop"
Environment
location = "A"
rooms = {
"A": "Dirty",
"B": "Dirty"
}
print("GOAL-BASED VACUUM CLEANER")
print("------------------------")
while True:
print(f"\nLocation : {location}")
print(f"Room A : {rooms['A']}")
print(f"Room B : {rooms['B']}")
action = goal_based_agent(location, rooms)
print(f"Action : {action}")
if action == "Suck":
rooms[location] = "Clean"
elif action == "Move Right":
location = "B"
elif action == "Move Left":
location = "A"
elif action == "Stop":
print("\nGoal achieved: Both rooms are clean!")
break

```
PS C:\Users\BMSCE\Desktop\AIVS>  c:; cd 'c:\Users\BMSCE\Desktop\AIVS'; & 'c:\python314\python.exe' 'c:\Users\BMSCE\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '58213' '--' 'c:\Users\BMSCE\Desktop\AIVS\vaccumgoal.py'
GOAL-BASED VACUUM CLEANER
------------------------

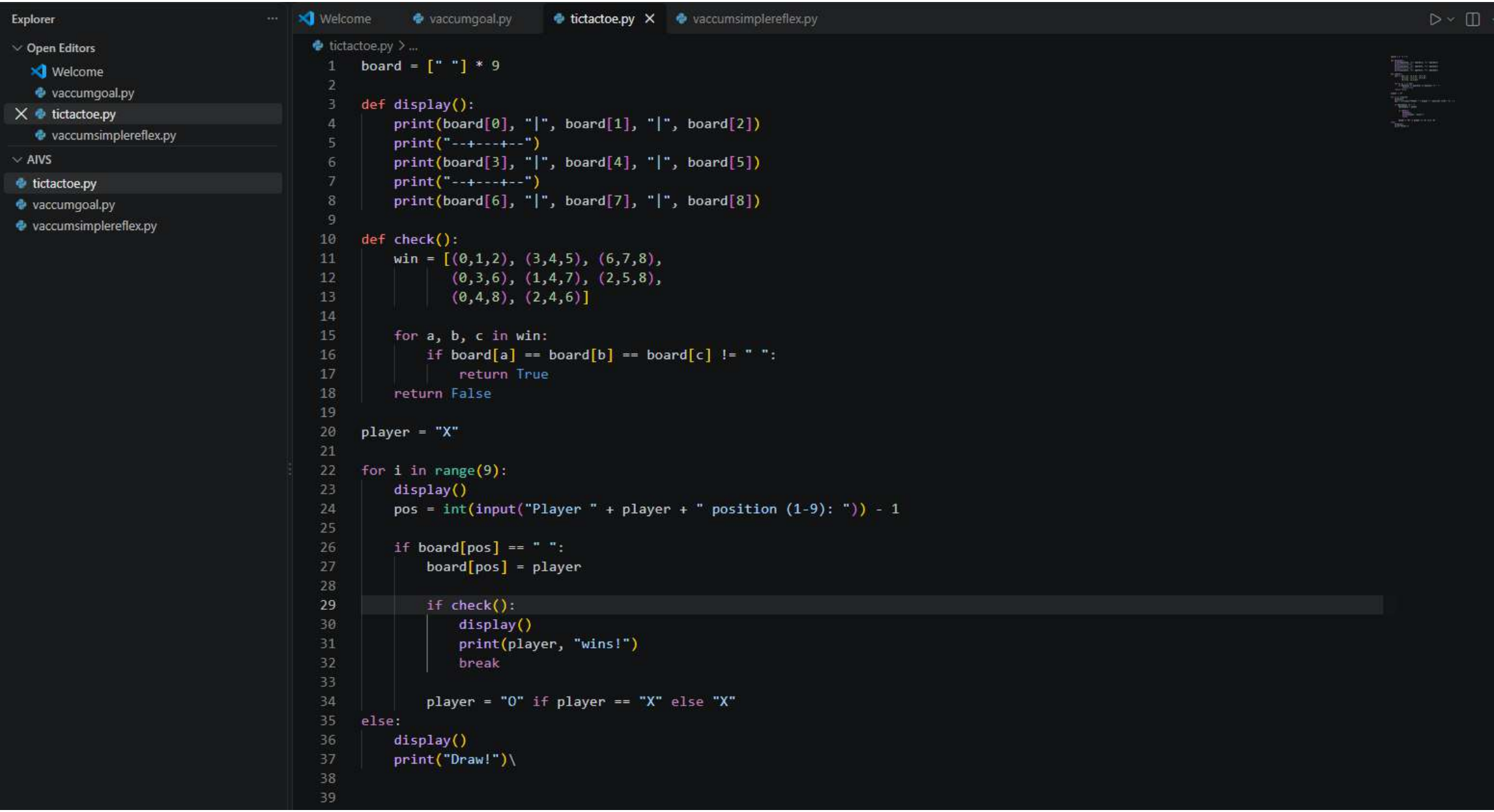
Location : A
Room A   : Dirty
Room B   : Dirty
Action   : Suck

Location : A
Room A   : Clean
Room B   : Dirty
Action   : Move Right

Location : B
Room A   : Clean
Room B   : Dirty
Action   : Suck

Location : B
Room A   : Clean
Room B   : Clean
Action   : Stop

Goal achieved: Both rooms are clean!
PS C:\Users\BMSCE\Desktop\AIVS>
```


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Welcome
vaccumgoal.py
tictactoe.py
vaccumsimplereflex.py
tictactoe.py > ...
board = [" "] * 9
def display():
print(board[0], "|", board[1], "|", board[2])
print("--+---+--")
print(board[3], "|", board[4], "|", board[5])
print("--+---+--")
print(board[6], "|", board[7], "|", board[8])
def check():
win = [(0,1,2), (3,4,5), (6,7,8),
(0,3,6), (1,4,7), (2,5,8),
(0,4,8), (2,4,6)]
for a, b, c in win:
if board[a] == board[b] == board[c] != " ":
return True
return False
player = "X"
for i in range(9):
display()
pos = int(input("Player " + player + " position (1-9): ")) - 1
if board[pos] == " ":
board[pos] = player
if check():
display()
print(player, "wins!")
break
player = "O" if player == "X" else "X"
else:
display()
print("Draw!")\

```
Goal achieved: Both rooms are clean!
PS C:\Users\BMSCE\Desktop\AIVS>  c:; cd 'c:\Users\BMSCE\Desktop\AIVS'; & 'c:\python314\python.exe' 'c:\Users\BMSCE\.vscode\extensions\ms-python.debugpy-2026.6.
0-win32-x64\bundled\libs\debugpy\launcher' '64282' '--' 'c:\Users\BMSCE\Desktop\AIVS\tictactoe.py'
  |   |
--+---+--
  |   |
--+---+--
  |   |
Player X position (1-9): 1
X |   |
--+---+--
  |   |
--+---+--
  |   |
Player O position (1-9): 6
X |   |
--+---+--
  |   | O
--+---+--
  |   |
Player X position (1-9): 2
X | X |
--+---+--
  |   | O
--+---+--
  |   |
Player O position (1-9): 5
X | X |
--+---+--
  | O | O
--+---+--
  |   |
Player X position (1-9): 3
X | X | X
--+---+--
  | O | O
--+---+--
  |   |
X wins!
PS C:\Users\BMSCE\Desktop\AIVS>
```